

FRANCISCO LOPES

Software Engineering · Agentic AI & Distributed Systems

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PROFILE

Master's student in Distributed Systems (University of Minho, Braga) with a rare combination of blockchain development experience and hands-on developer tooling work. Built and shipped a code intelligence tool at Neadvance that was adopted by the engineering team. Independently developed a full-stack decentralised voting application in Solidity and React, giving me direct domain knowledge of the on-chain systems Uphold is built on. Currently engineering a production-grade distributed IoT telemetry platform using ZMQ, Erlang, and Java. I learn fast in codebases I've never seen before, ship working software under real constraints, and care about the problems engineers face daily.

EXPERIENCE

Software Engineering Intern · Neadvance

Aug 2024 – Sep 2024 Braga, Portugal

- Identified a workflow bottleneck where large Git pushes were difficult to review, and independently designed and built a Git diff visualisation tool from scratch using Python
- Developed a clean interface that parses and displays code changes with clear indentation, highlighting added and removed lines for rapid review
- Delivered a measurable improvement in the team's code review efficiency, with the tool adopted and continued in active use after the internship ended

EDUCATION

MSc Software Engineering — Distributed Systems · University of Minho

Sep 2025 – Present Braga, Portugal

BSc Software Engineering · University of Trás-os-Montes and Alto Douro

Sep 2022 – Jun 2025 Vila Real, Portugal

PROJECTS

Distributed IoT Telemetry Platform · MSc Distributed Systems — University of Minho

Jan 2026 – Present

- Designed and implemented concurrent message handling using Java NIO and RxJava reactive streams; ZMQ DEALER/ROUTER for scalable async inter-server communication
- Gained deep practical understanding of distributed coordination, message-passing protocols, and fault-tolerant system design

Votium — Blockchain Voting Application · BSc Final Project

Feb 2025 – Jul 2025 Grade: 17/20

- Architected and deployed smart contracts in Solidity on an Ethereum-compatible (Ganache) network, designed the on-chain data structures for vote storage and verification
- Built the full frontend in React and JavaScript with real-time blockchain event listening, giving me end-to-end ownership of a Web3 product stack

SKILLS & TECHNOLOGIES Light involvement

Languages: Python · Java · JavaScript · TypeScript · Solidity · Erlang

Frameworks & Tools: React · RxJava · Java NIO · ZeroMQ · Git · Ganache · Node.js

Domains: Blockchain development · Distributed systems · Developer tooling · Code intelligence

LANGUAGES

Portuguese (Native) · **English** (Proficient — written and oral) · **Spanish** (Intermediate) · **German** (Beginner)